Dr. Henrik Wilhelm Edler

Contact & personal information

ASTRON
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Citizenship: Germany
Languages: English (proficient), German (native), Dutch (B1)

Research interest Galaxy evolution, galaxy clusters, non-thermal processes, star–planet interactions, radio surveys, radio interferometric techniques

Education

09/2020 – 01/2024 **PhD, Physics (Grade: 1.0 summa cum laude)**
Hamburg Observatory, University of Hamburg, Germany
Thesis title: "The Effect of the Galaxy Cluster Environment on Galaxies and AGN — A LOFAR Study"
Advisors: Prof. Dr. Marcus Brüggen, Prof. Dr. Francesco de Gasperin

04/2018 – 08/2020 **MSc, Physics (Grade: 1.0 with distinction)**
University of Hamburg, Hamburg, Germany
Thesis title: "A Comprehensive Simulation Tool for Low-Frequency Radio Observations with LOFAR"
Advisor: Prof. Dr. Francesco de Gasperin

04/2015 – 09/2018 **BSc, Physics (Grade: 1.3)**
University of Hamburg, Hamburg, Germany
Thesis title: "Indirect Detection of Dark Matter in Dwarf Galaxies with TAIGA"
Advisor: Prof. Dr. Dieter Horns

Professional Appointments

since 09/2024 **Bell Burnell Fellow** in the SKA Science Group
ASTRON, Netherlands Institute for Radio Astronomy, Dwingeloo, The Netherlands

01/2024 – 07/2024 **Postdoctoral Researcher** in the group of Prof. Marcus Brüggen
Hamburg Observatory, University of Hamburg, Germany

09/2020 – 01/2024 **Doctoral Researcher** in the group of Prof. Francesco de Gasperin
Hamburg Observatory, University of Hamburg, Germany

Publications in refereed journals

35 refereed publications since 2020, with a total of >1000 citations and an h-index of 15 (**6 as first author**, 1 as second author by an advised student, 5 more as second author). A complete publication list is enclosed below and available on NASA/ADS.

Grants & Third Party Funding

2026 **Lorentz Center Workshop** "The next frontier: Sub-arcsecond astronomy below 100 MHz" (25 000 € in-kind)
Lorentz Center, Leiden; main organizer; workshop taking place 04/2027

2025 **Dutch SKA Regional Center prototype, compute grant** (200 000 CPU h)
LOFAR LBA VLBI processing; .

2025 **DFG Research Grant 558370136** (280 000 €)
German Research Foundation (DFG); Co-I with Prof. Dr. M. Brüggen, lead the research plan

Awards, Honors & Fellowships

2024	**Bell Burnell Fellowship** (3+1-year independent fellowship), ASTRON
2024	**Stargazer Award** for the PhD thesis (5 000 €); Cologne, Germany
01/2024	**PhD with distinction** (summa cum laude), University of Hamburg
08/2020	**MSc with distinction**, University of Hamburg

Approved telescope proposals

since 2021 PI on 7 successful observing proposals across 4 facilities (LOFAR, MeerKAT, MWA, NenuFAR), and Co-I on >2400 h of approved LOFAR, MeerKAT and uGMRT time.

Proposals as Principal Investigator

under review	**XMM-Newton** (16 ks)
2026	**NenuFAR** (286 h; RP9A)
	"Monitoring the LOFAR-detected star–planet interaction candidate system GJ 725A"
2026	**MWA** (35 h; OA012)
	"Deep low-frequency observations of the Norma cluster"
2025	**MeerKAT** (19 h; SCI-20251101-HE-01)
	"Extreme Ram Pressure Stripping of the Virgo Cluster Dwarf Galaxy IC 3476"
2024	**MeerKAT** (8 h; SCI-20241101-HE-01)
	"Environmental Effects in the Merging Subcluster M49"
2024	**LOFAR+NenuFAR** (12 h; COM20_007 – commissioning)
	"Commissioning of the NenuFAR-LOFAR Super Station mode"
2022 – 2023	**LOFAR** (60 h; LC18_012, LC20_038)
	"The LOFAR LBA Virgo Cluster Survey"

Selected Proposals as Co-Investigator

2026	**uGMRT** (22 h; 50_005; Co-I)
	"Nearby star-forming galaxies as laboratories of magnetic fields and cosmic ray transport"
2025	**MeerKAT** (38 h; SCI-20251102-IR-01; Co-I & technical lead)
	"MAUVE-MeerKAT: A pilot S-band survey of Virgo cluster galaxies"
2022 – 2023	**MeerKAT** (285 h; MKT-22008, MKT-23067; Co-I, lead the survey strategy and the pilot campaign)
	"The MeerKAT Virgo Cluster Survey"
2023	**LOFAR** (118 h; LC17_006, LC20_011; Co-I)
	"Radio Megahalos in Galaxy Clusters"
2023	**LOFAR** (180 h; LC20_005; Co-I)
	"LoLSS-Deep"
2022	**uGMRT** (35 h; 43_035; Co-I)
	"Re-energetisation of fossil plasma in clusters"
2021	**LOFAR** (1800 h; LT16_004; Co-I)
	"The LOFAR LBA Sky Survey (LoLSS)"

Collaboration Membership

SKA Extragalactic Continuum Science Working Group
LOFAR Surveys Key Science Project
ViCTORIA — Virgo Cluster multi-Telescope Observations in Radio of Interacting galaxies and AGN
MAUVE — MeerKAT/MUSE/ALMA survey of Virgo cluster galaxies
LOFAR 2.0 Large Programmes: LOFAR LBA Community Sky Survey (LLoCuSS), Nearby Galaxies, Exoplanet Science with LOFAR (ExLOO),

Supervision experience

since 09/2026	**A. Mahesh** — MSc student, U. Groningen	daily supervisor
06/2026 – 08/2026	**M. Brown** — summer student, ASTRON	secondary supervisor
04/2026 – 07/2026	**S. van Schoot** — BSc student, U. Groningen	secondary supervisor
since 02/2026	**D. Fistos** — MSc student, U. Leiden	primary supervisor
06/2025 – 08/2025	**L. H. Long** — summer student, ASTRON	primary supervisor
04/2023 – 03/2024	**A. Spasic** — MSc student, U. Hamburg	daily supervisor
09/2021 – 08/2022	**T. Beeth** — MSc student, U. Hamburg	daily supervisor

Teaching

08/2026	**Lecture** — Galaxies and Galaxy Clusters ASTRON summer student program
10/2025	**Lecture** — Introduction to Radio Astronomy ASTRON onboarding program
03/2021	**Lecturer** — LOFAR Data School 2021 (two sessions incl. hands-on training) virtual
04/2020 – 08/2020	**Teaching Assistant** — Introduction to Astronomy (BSc level) University of Hamburg
04/2017 – 08/2017	**Student Tutor** — Electrodynamics and Optics (BSc level) University of Hamburg

Scientific software development

since 2020	**Co-developer of LiLF** (Library for Low Frequencies; github) – LOFAR LBA calibration pipeline
since 2019	**Co-developer of LoSiTo** (LOFAR 2.0 Simulation Tool; github)

Talks invited

	"Unveiling environmental interactions in nearby galaxy clusters in the radio continuum"
10/2026	**SPARCS 2026**, Seoul, Korea "The LOFAR LBA Sky Survey – Second data release and recent results"
11/2025	**Radio2025**, MPIA Heidelberg, Germany "Wide-field VLBI with the LOFAR LBA"
11/2025	**SPARCS 2025**, virtual "Wide-field VLBI with the LOFAR LBA"
11/2025	**Harvard CfA galaxy cluster group meeting**, virtual "The Radio View of the Extreme Ram Pressure Stripping in the Abell 1367 Cluster"
09/2025	**LOFAR Family Meeting 2025**, Paris, France "Galaxy clusters with LOFAR LBA VLBI"
02/2025	**LOFAR Imaging Forum 2025**, virtual "LOFAR Decametric Imaging"
12/2024	**SALF 2024**, Shanghai, China "The LOFAR View of Ram Pressure Stripping in Nearby Galaxy Clusters"
11/2023	**GLOW & Radio2023**, Bochum, Germany "The LOFAR View of Ram-Pressure Stripping in the Virgo Cluster"
03/2023	**Harvard CfA galaxy cluster group meeting**, virtual "The LOFAR View of Ram-Pressure Stripping in the Virgo Cluster"
03/2022	**Clusters & Relics Workshop**, TLS Tautenburg, Germany "Abell 1033 — Ultra Steep-Spectrum Emission at Ultra-low Frequencies"

Service

Service to Profession

04/2027 (upcoming) **Primary organizer and initiator**, Lorentz Center Workshop "The next frontier: Sub-arcsecond astronomy below 100 MHz", Leiden, The Netherlands

05/2026 **SOC member**, Netherlands Astronomy Conference, Egmond aan Zee, The Netherlands

05/2026 **Organizer**, ASTRON/Kapteyn Institute postdoc training: "Preparing strong grant applications", Groningen, The Netherlands

2025 **LOC member**, conference "Beyond Earth: Lunar and Space Radio Missions", Dwingeloo, The Netherlands

2023 **Organizing committee**, LOFAR Early-Career Session, Olsztyn, Poland
Conference session for early-career researchers, covering grant and observing proposals

2023 **Organizing committee**, LOFAR Early-Career Event, Leiden, The Netherlands
Three-day conference with science talks and ECR-specific workshops

Referee Astronomy & Astrophysics, The Astrophysical Journal, Publications of the Astronomical Society of Australia, Journal of Astronomical Instrumentation

Institutional Service

since 07/2026 **Postdoc representative** (ASTRON)

since 03/2025 Organizer of the weekly **Astrolunch talks** (ASTRON)

Public Outreach

07/2025 **TV report** — Presentation of the LOFAR telescope and my research in a report by the German public broadcaster NDR (NDR media library)

2024 **Open days** — Support for the open day at ASTRON

2022 – 2023 **Open days** — Support for the open days at the Hamburg Observatory

2021 – 2024 **Public tours** — Guide for public tours and support for the open days at the Hamburg Observatory

Publication List refereed journals

First-author publications

[1] **H. W. Edler**, J. R. Callingham, I. Davis, C. Cordun, F. de Gasperin, H. L. Le, B. J. S. Pope, H. K. Vedantham, R. J. van Weeren, and P. Zarka. "Coherent radio emission at 45 MHz from the solar neighborhood planet-hosting star GJ 725A" (2026). **Submitted for publication in A&A**.

[2] **H. W. Edler**, M. Arias, A. Botteon, and C. G. Bassa. "Multiwavelength insights into the pulsar wind nebula candidate near 1LHAASO J0343+5254u: An obscured merging galaxy cluster?" *A&A* 708 (2026), A154.

[3] **H. W. Edler**, M. Hoeft, S. Bhagat, et al. "The longest known tails of ram-pressure-stripped star-forming galaxies are caused by an intracluster medium shock in Abell 1367". *A&A* 706 (2026), A240.

[4] **H. W. Edler**, I. D. Roberts, A. Boselli, F. de Gasperin, V. Heesen, M. Brüggen, A. Ignesti, and L. Gajović. "ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies". *A&A* 683 (2024), A149.

[5] **H. W. Edler**, F. de Gasperin, T. W. Shimwell, et al. "VICTORIA project: The LOFAR HBA Virgo Cluster Survey". *A&A* 676 (2023), A24.

[6] **H. W. Edler**, F. de Gasperin, G. Brunetti, A. Botteon, V. Cuciti, R. J. van Weeren, R. Cassano, T. W. Shimwell, M. Brüggen, and A. Drabent. "Abell 1033: Radio halo and gently reenergized tail at 54 MHz". *A&A* 666 (2022), A3.

[7] **H. W. Edler**, F. de Gasperin, and D. Rafferty. "Investigating ionospheric calibration for LOFAR 2.0 with simulated observations". *A&A* 652 (2021), A37.

Publications with major contributions

[1] V. Heesen, **H. W. Edler**, M. Brüggen, M. Stein, D. J. Bomans, R. Paladino, K. T. Chyży, K. Małek, M. A. Lara-López, and F. S. Tabatabaei. "Nearby galaxies in the LOFAR Two-metre Sky Survey: IV. A fundamental plane of the radio-star formation rate relation". *A&A* 712 (2026), A242.

[2] T. Vičánek Martínez, **H. W. Edler**, and M. Brüggen. "Simulating realistic radio continuum survey maps with diffusion models". *A&A* 700 (2025), A18.

[3] F. de Gasperin, **H. W. Edler**, A. Boselli, et al. "The ViCTORIA project: Description of a multi-frequency radio survey of the Virgo galaxy cluster". *A&A* 693 (2025), A189.

[4] A. Spasic, **H. W. Edler**, Y. Su, et al. "ViCTORIA project: The LOFAR-MeerKAT view of active galactic nuclei in Virgo cluster early-type galaxies". *A&A* 690 (2024), A195.

[5] F. de Gasperin, **H. W. Edler**, W. L. Williams, et al. "The LOFAR LBA Sky Survey. II. First data release". *A&A* 673 (2023), A165.

[6] T. Pasini, **H. W. Edler**, M. Brüggen, et al. "Particle re-acceleration and diffuse radio sources in the galaxy cluster Abell 1550". *A&A* 663 (2022), A105.

Co-author publications

[1] A. Boselli, A. Lupi, P. Serra, et al. "A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE): XX. Star formation in the tidal tail of NGC 4254". *A&A* 712 (2026), A45.

[2] C. Russeeawon, J. S. Kenyon, H. L. Bester, **H. W. Edler**, F. de Gasperin, and O. M. Smirnov. "Robust clock and ionospheric phase delay separation in radio interferometric calibration". *MNRAS* 549.4 (2026).

[3] F. de Gasperin, M. Brüggen, **H. Edler**, R. J. van Weeren, T. Venturi, and G. Brunetti. "The Cosmic Ray Life Cycle in Galaxy Clusters". *arXiv e-prints* (2026), arXiv:2606.26005.

[4] A. Ignesti, I. D. Roberts, **H. W. Edler**, M. Gitti, R. J. van Weeren, P. Serra, and P. Jachym. "Exploring the physics of ram pressure stripping with radio continuum observations in the SKA era". *arXiv e-prints* (2026), arXiv:2606.25746.

[5] T. W. Shimwell, M. J. Hardcastle, C. Tasse, et al. "The LOFAR Two-metre Sky Survey: VII. Third Data Release". *A&A* 707 (2026), A198.

[6] J. M. Boxelaar, F. De Gasperin, M. J. Hardcastle, J. H. Croston, L. K. Morabito, R. J. van Weeren, and **H. Edler**. "LOFAR 58 MHz Legacy Survey of the 3CRR catalog". *A&A* 704 (2025), A65.

[7] G. Lusetti, M. Brüggen, **H. W. Edler**, et al. "A view of the CIZA J2242.8+5301 galaxy cluster at very low radio frequencies". *A&A* 699 (2025), A200.

[8] L. Gajović, V. Heesen, M. Brüggen, et al. "The low-frequency flattening of the radio spectrum of giant H II regions in M 101". *A&A* 695 (2025), A41.

[9] L. Gajović, B. Adebahr, A. Basu, et al. "A spatially resolved radio spectral study of the galaxy M 51". *A&A* 689 (2024), A68.

[10] H. McCall, T. H. Reiprich, A. Veronica, et al. "The SRG/eROSITA All-Sky Survey: View of the Virgo Cluster". *A&A* 689 (2024), A113.

[11] T. Pasini, F. De Gasperin, M. Brüggen, et al. "Ultra-low-frequency LOFAR spectral indices of cluster radio halos". *A&A* 689 (2024), A218.

[12] I. D. Roberts, R. J. van Weeren, F. de Gasperin, A. Botteon, **H. W. Edler**, A. Ignesti, L. Matijević, and N. Tomičić. "A 100 kpc ram pressure tail trailing the group galaxy NGC 2276". *A&A* 689 (2024), A22.

[13] V. Heesen, S. Schulz, M. Brüggen, **H. W. Edler**, M. Stein, R. Paladino, A. Boselli, A. Ignesti, M. Fossati, and R.-J. Dettmar. "Nearby galaxies in the LOFAR Two-metre Sky Survey. III. Influence of cosmic-ray transport on the radio-SFR relation". *A&A* 682 (2024), A83.

[14] G. Lusetti, F. de Gasperin, V. Cuciti, et al. "Re-energization of AGN head-tail radio galaxies in the galaxy cluster ZwCl 0634.1+47474". *MNRAS* 528.1 (2024), pp. 141–159.

[15] F. Geyer, K. Schmidt, J. Kummer, M. Brüggen, **H. W. Edler**, D. Elsässer, F. Griese, A. Poggenpohl, L. Rustige, and W. Rhode. "Deep-learning-based radiointerferometric imaging with GAN-aided training". *A&A* 677 (2023), A167.

[16] A. Boselli, P. Serra, F. de Gasperin, et al. "ViCTORIA project: MeerKAT H I observations of the ram pressure stripped galaxy NGC 4523". *A&A* 676 (2023), A92.

[17] V. Cuciti, F. de Gasperin, M. Brüggen, et al. "Galaxy clusters enveloped by vast volumes of relativistic electrons". *Nature* 609.7929 (2022), pp. 911–914.

[18] T. W. Shimwell, M. J. Hardcastle, C. Tasse, et al. "The LOFAR Two-metre Sky Survey. V. Second data release". *A&A* 659 (2022), A1.

[19] N. Biava, F. de Gasperin, A. Bonafede, et al. "The ultra-steep diffuse radio emission observed in the cool-core cluster RX J1720.1+2638 with LOFAR at 54 MHz". *MNRAS* 508.3 (2021), pp. 3995–4007.

[20] F. de Gasperin, W. L. Williams, P. Best, et al. "The LOFAR LBA Sky Survey. I. Survey description and preliminary data release". *A&A* 648 (2021), A104.

[21] F. de Gasperin, G. Brunetti, M. Brüggen, et al. "Reaching thermal noise at ultra-low radio frequencies. Toothbrush radio relic downstream of the shock front". *A&A* 642 (2020), A85.